Virtual University of Pakistan

Bt301

Introduction to biotechnology

Midterm past papers solved

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Mcq

1...transfers phosphate from atp
2..... Has single chromosome with protein core
3Dna has one oh removed at .... Carbon
4 Pcr is used for ....
Dna rna dna$rna protein
5 .... Amplifies the number in vitro
Pcr hybridization clonning
6 Hae 3 is identifies .... Nucleotides
7 Ti is found in .....
8 Claudia wala aya tha k new ka nam kya tha
9 Making of rna from dna is called .....
10 Yeast genome was sequenced till.....
11 Nomenclature was presented by.....
12 Dna is lysed by..... Enzymes
13 Psi angle of a helix is.....

Q. Xray difrction 5mrks

Ans: Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibers. They showed that DNA produces a characteristic x-ray diffraction pattern.

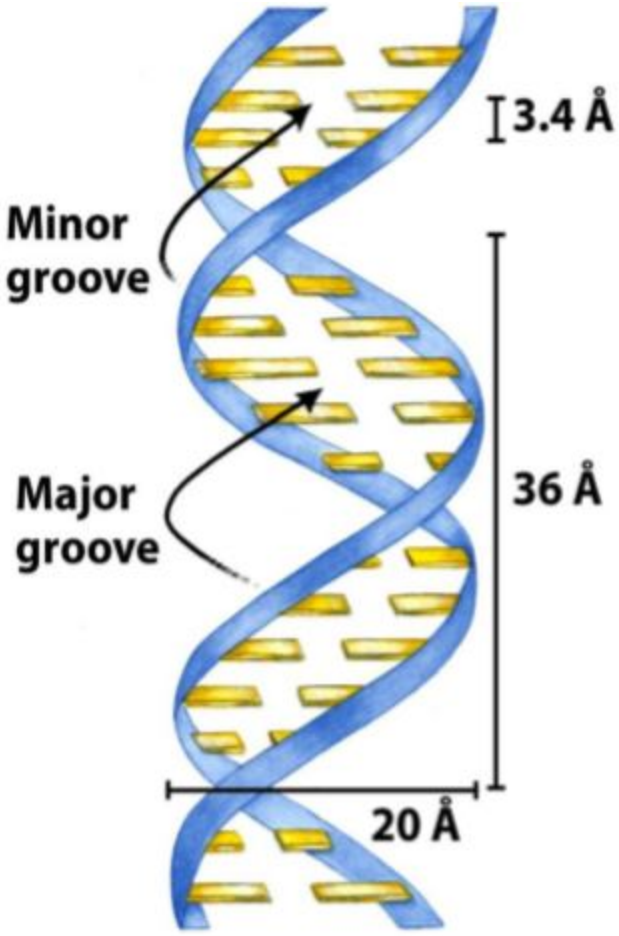
Q. Electroporation 5

Ans: Electroporation

§ A rapid and simple technique for introducing cloned genes into a wide variety of microbial, plant and animal cell, including E. coli is electroporation § In electroporation, electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

Q.Minor and major groove length and nucleotides 3

Ans: The pairing and coiling of the two strands create a major groove and minor groove on the surface of the helix.



Q. Gene 2

Ans: A piece of DNA (or RNA) that contains the primary sequence to produce a functional biological gene product (RNA or protein).

Q. branches of biotechnology?

Ans: White Biotechnology Development of processes and microorganisms for Industrial processes. Key concept Scale of operation Low volume high price

White Biotechnology Example • Enzyme production

Red Biotechnology It is concerned with the discovery and development of innovative drugs and treatments.

Green Biotechnology It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones. Green Biotechnology Example Bt corn from Syngenta Bt cotton from Monsanto

Q. ori region

Ans: Plasmids encode only a few of the proteins required for their own replication § Host range of plasmid is determined by ori region § Plasmids whose ori

derived from Col E1 have restricted host range

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